

Prashant Kafle

+977-9741897738 | prashantkafle7738@gmail.com | linkedin.com/in/prashantkafle33 | github.com/prashant7738 | prashantkafle33.com.np

SUMMARY

AI Engineer and Computer Engineering student with production experience building RAG pipelines, multi-agent LLM systems, and AI-powered full-stack applications. Published research on evaluating AI agents with LLM-based judges. Proficient in LangGraph, LangChain, pgvector, Groq, HuggingFace, and FastAPI. Focused on reliable, cost-efficient LLM architectures that ship to real users.

EDUCATION

Thapathali Engineering College, IOE – Tribhuvan University
Bachelor of Engineering in Computer Engineering

Kathmandu, Nepal
Jan. 2024 – Jan. 2028

EXPERIENCE

Freelance AI & Full-Stack Developer
Self-Employed

Mar. 2025 – Present
Remote

- Integrated Groq and OpenAI APIs into production apps; designed prompt pipelines with fallback strategies and RAG-based document search using pgvector on PostgreSQL
- Delivered full-stack applications with React, Django REST Framework, JWT auth, and RBAC; maintained documentation for non-technical clients across remote workflows

PROJECTS

- Contextra AI** | *FastAPI, RAG, pgvector, PostgreSQL, BGE Embeddings, HuggingFace, Astro, React* | [Live](#) 2026
- Architected a production RAG system for PDF Q&A; hybrid OCR pipeline (PyMuPDF + EasyOCR) feeds BGE-small embeddings into pgvector for grounded, citation-backed LLM answers
 - Added Pareto 80/20 summary and smart flashcard generation features; JWT multi-workspace auth; FastAPI backend with SQLAlchemy + Alembic
- Multiagent Data Analyst** | *Python, LangGraph, Groq API, Llama 3.3-70B, Pandas, NumPy, Matplotlib* 2026
- Built a LangGraph DAG orchestrating 4 stateless agents (Structural Profiler → Semantic Tagger → Preprocessor → Statistical Analyser) to process raw CSV files end-to-end
 - Semantic Tagger uses Groq Llama 3.3-70B with minimal-token prompts; Preprocessor runs a 10-step pipeline with IQR outlier clipping, normalisation, and a 0–100 data quality score
- HealthSaathi** | *React, Django, Groq LLM, GPT-4, ChromaDB, PostgreSQL, Leaflet.js* 2026
- AI medical triage app with bilingual (Nepali/English) symptom assessment; integrated Groq LLaMA + GPT-4 with ChromaDB semantic caching to cut redundant API calls; Hackathon winner
- Live Bidding NP** | *React, Django Channels, SQLAlchemy Core, PostgreSQL, WebSockets, Redis* 2025 – 2026
- Real-time auction platform with concurrent WebSocket bid handling via Django Channels + Redis; SQLAlchemy Core for raw-SQL performance; JWT in HttpOnly cookies against XSS

PUBLICATIONS

- Evaluating AI Agents with LLM-Based Judges** 2026
Prashant Kafle, Sunil Paudel *Undergraduate research paper*
- Compared ROUGE-L, Bradley-Terry, Elo-style ranking, and judge-human agreement analysis for evaluating agent responses on GAIA-style tasks
 - PDF available in portfolio and resume assets: [paper link](#)

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML/CSS, C/C++
AI & LLM: RAG, LangGraph, LangChain, Multi-Agent Systems, Prompt Engineering, HuggingFace, Groq API, OpenAI API, BGE Embeddings, pgvector, ChromaDB
Backend & APIs: FastAPI, Django, Django REST Framework, Django Channels, WebSockets, JWT Auth
Data & Databases: Pandas, NumPy, Matplotlib, Scikit-learn, PostgreSQL, SQLAlchemy Core, Redis
Languages & Tools: Python, JavaScript, TypeScript, SQL, React, Astro, Git, Docker, Linux